Predictive Structure of the Perpetual-Futures Limit Order Book

Universality, Orthogonality, and the Execution Boundary

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Abstract

Using a 236-feature representation of the limit order book computed at 100 ms cadence on BTC, ETH and SOL perpetual futures (2.17 million ticks per symbol), we document the predictive structure of the book at horizons of one second to one hundred minutes. Directional information is large (rank information coefficient up to 0.47 at 1–5 s), universal across symbols and volatility regimes, concentrated in eight independent feature axes led by order-book imbalance, and spectrally localized in the 0.005–0.1 Hz band with an Ornstein–Uhlenbeck half-life of 5–7 s. Volatility information (peak IC 0.35) resides in a disjoint feature family — arrival intensity, activity counts, toxicity — and the two channels are empirically orthogonal: directional features carry no volatility information and vice versa, on all three symbols. The signal decays toward efficiency in real time (its half-life shortening as spreads tighten) and is sharply bounded at execution: conditioning on directionally-consistent maker fills collapses the IC to ≈ 0.03, while conditioning on the presence of any fill *raises* it to 0.52, isolating adverse selection — rather than signal decay — as the binding constraint. An out-of-time replication on an independent month of data is in progress. We discuss implications: two-block (sign $\times$ size) forecasting architectures, entropy-gated conditioning, and execution-timing overlays for slower strategies.

Keywords: limit order book, market microstructure, cryptocurrency, perpetual futures, information coefficient, adverse selection, spectral analysis

JEL Classification: C58, G12, G14

Contents

*Working paper, preliminary draft — comments welcome. The out-of-time replication study (Section 9) is in progress; all other estimates are from the internal empirical record consolidated 2026-07-25. Please do not cite specific numbers without checking for a newer version.

1 Introduction

That order-book imbalance predicts short-horizon price moves is well established: Cont et al. (2014) document the linear price impact of order-book events in equities, and a large subsequent literature confirms the result across venues and with increasingly flexible models. This paper does not re-claim that finding. Instead, we use a purpose-built measurement platform — 236 features computed every 100 ms from the full depth, trade flow, and funding state of three cryptocurrency perpetual-futures books — to establish four facts that are, to our knowledge, new.

1. **Orthogonality (the factorization result).** The directional and volatility channels of the book are carried by *disjoint* feature families. Imbalance-family features predict the sign of returns (IC up to 0.47) and carry zero volatility information; intensity-family features predict magnitude (IC up to 0.35) and carry zero directional information. This holds feature by feature, on all three symbols. Sign and size of short-horizon returns are independently predictable — a factorization with direct architectural consequences (Section 5).

2. **Universality at scale.** A census of 207 live features across six horizons and three symbols, with bootstrap confidence intervals, compresses to eight independent directional axes whose IC ordering is preserved across symbols (Section 3).

3. **Spectral localization and real-time efficiency drift.** The entire directional IC lives in the 0.005–0.1 Hz band; the imbalance process has an Ornstein–Uhlenbeck half-life of 5–7 s; and between two measurement windows six weeks apart the 5 s IC decayed from 0.45 to 0.29 while the 1 s IC held — the signal's half-life is shortening as the venue's spreads tighten (Section 6).

4. **The execution boundary.** Conditioning the IC on directionally-consistent maker fills collapses it from 0.45 to ≈ 0.03; conditioning on *any* fill raises it to 0.52. The decomposition locates the loss at fill selection, not at signal decay, and isolates adverse selection in the sense of Glosten and Milgrom (1985) as the binding mechanism (Section 8). Regime conditioning on a book-shape entropy measure recovers part of the boundary (+22% IC in the low-entropy quintile).

The perspective throughout is measurement-first: we report what the book knows, where in frequency it knows it, how the two channels factorize, and precisely where the information stops being capturable. Section 9 pre-registers an out-of-time replication of every estimate on an independent month of data. Section 10 discusses implications for forecasting architecture and execution design.

2 Data and Measurement Platform

Venue and instruments. We study BTC, ETH and SOL perpetual futures on Hyperliquid, a high-throughput decentralized derivatives venue. Perpetual futures are the dominant crypto trading instrument and carry a funding-rate mechanism whose pricing theory is developed in Ackerer et al. (2025).

Feature computation. A Rust ingestion engine subscribes to the full order-book and trade stream and emits a 236-dimensional feature vector per symbol every 100 ms, spanning 21 categories: book imbalance at multiple depths, raw depth, microstructure dynamics (queue position, imbalance velocity), trade flow and aggressor ratios, VWAP deviation, entropy measures (permutation entropy, book-shape entropy), Hawkes-type arrival intensity, realized and range-based volatility, spread and toxicity proxies, funding/open-interest state, and cross-symbol imbalance aggregates. End-to-end feature latency is below 80 ms at the 99th percentile.

Census window. The headline census uses three contiguous days (2026-05-19 to 2026-05-21), 2.17 million ticks per symbol at 100 ms cadence. Rank (Spearman) ICs against future mid-price returns are computed at six horizons (1 s, 5 s, 30 s, 1 m, 5 m, 15 m; selected features also at 100 m) on ∼21,708 subsampled points per symbol (one per ∼10 s to control overlap), with significance at $p < 0.01$ and multiplicity handled by Benjamini–Hochberg false-discovery control (Benjamini and Hochberg, 1995). Of 236 computed features, 207 were live in the census window. The fill-conditioning study (Section 8) uses a longer window: 31 calendar dates (2026-04-19 to 2026-06-04), of which 23 passed data-quality gates.

Honest accounting. Two limitations are flagged here rather than discovered later. First, the census window is short; temporal out-of-sample stability initially *failed* our own validation matrix (Section 7), which is precisely what the replication study addresses. Second, the platform's data-continuity record over April–June contains gaps; all estimates are computed on gap-audited clean days only.

3 The Directional Channel

Table 1 reports peak rank ICs for the leading directional features. Order-book imbalance dominates: at 1–5 s horizons the level-1 quantity imbalance reaches IC +0.45 to +0.47 depending on symbol, with the level-5 and notional variants essentially tied. The ordering of features is preserved across all three symbols — the hierarchy is structural, not asset-specific.

Eight independent axes. The ∼29 fast-directional features are heavily cross-correlated. They compress into eight axes with approximate standalone IC: (1) book imbalance ∼0.45; (2) raw depth asymmetry ∼0.42; (3) cross-symbol imbalance ∼0.35; (4) queue dynamics ∼0.31; (5) VWAP deviation, mean-reverting ∼0.25; (6) imbalance velocity ∼0.19; (7) imbalance entropy ∼0.17; (8) aggressor flow ∼0.11.

Decay. Table 2 traces the peak directional IC across horizons. The half-life is ∼30 s for BTC, ∼20 s for ETH and ∼15 s for SOL — the thinner the book, the faster the decay — and by five minutes roughly 80% of the signal is gone. A distinct *slow* family grows with horizon instead (e.g. a 1-hour divergence measure reaching IC 0.21 at 100 m), but its analysis is outside this paper's scope.

Table 1: Leading directional features: peak rank IC by symbol (census window 2026-05-19/21; horizons 1–5 s unless noted).

Feature	BTC	ETH	SOL	Axis
`imbalance_qty_l1`	+0.453	+0.447	+0.466	book imbalance
`imbalance_qty_l5`	+0.456	+0.431	+0.453	book imbalance
`imbalance_notional_l5`	+0.456	+0.431	+0.453	book imbalance
`imbalance_orders_l5`	+0.435	+0.438	+0.455	book imbalance
`imbalance_depth_weighted`	+0.434	+0.410	+0.425	book imbalance
`raw_bid_depth_5`	+0.420	+0.405	+0.406	depth asymmetry
`raw_ask_depth_5`	−0.424	−0.404	−0.415	depth asymmetry
`cross_obi_mean`	+0.354	+0.334	+0.331	cross-symbol
`micro_queue_position_bid`	+0.307	+0.282	+0.291	queue dynamics
`flow_vwap_deviation` (1 s)	−0.287	−0.206	−0.188	VWAP reversion
`micro_obi_velocity` (1 s)	+0.192	+0.181	+0.208	imbalance momentum
`ent_permutation_imbalance_16` (1 s)	−0.170	−0.173	−0.177	imbalance entropy
`flow_aggressor_ratio_5s`	+0.112	+0.109	+0.111	aggressor flow

Table 2: Peak directional IC by horizon (census window).

Symbol	1 s	5 s	30 s	1 m	5 m	15 m
BTC	0.447	0.456	0.261	0.187	0.090	0.056
ETH	0.447	0.438	0.219	0.160	0.074	0.031
SOL	0.466	0.412	0.190	0.135	0.056	0.036

4 The Volatility Channel

Table 3 reports the leading predictors of short-horizon volatility (absolute-return magnitude) at 5 s. The family is entirely different in character: self-excitation of trade arrivals (a Hawkes-type intensity), realized and range-based volatility over trailing windows, raw activity counts, and an effective-spread toxicity proxy in the spirit of the flow-toxicity literature (Easley et al., 2012).

Table 3: Leading volatility features: rank IC vs. future absolute return at 5 s (census window).

Feature	BTC	ETH	SOL
`hawkes_intensity`	+0.345	+0.281	+0.247
`vol_returns_5m`	+0.328	+0.326	+0.293
`vol_parkinson_5m`	+0.318	+0.312	+0.281
`flow_count_30s`	+0.318	+0.281	+0.253
`illiq_trade_count`	+0.318	+0.286	+0.261
`toxic_effective_spread`	+0.291	+0.279	+0.254

Notably, VPIN-style measures and tick-sequence entropy carry magnitude information only — they have no directional IC at any horizon tested, on any symbol.

5 Orthogonality: The Factorization Result

The paper's central positive finding is the clean separation between the two channels. Every feature in Table 1 has volatility-IC indistinguishable from zero, and every feature in Table 3 has directional IC indistinguishable from zero — consistently across BTC, ETH and SOL. Direction and volatility of short-horizon returns are predictable *from disjoint measurements of the same order book.*

Three consequences follow. First, architecturally: a forecasting system can be built as two independent blocks — a sign block driven by the imbalance family and a size/veto block driven by the intensity family — with no interaction terms to estimate and hence no interaction terms to overfit. Second, econometrically: the factorization licenses separate estimation and separate validation of the two blocks, halving the effective multiple-testing burden. Third, economically: it suggests the two channels reflect different underlying processes — resting-liquidity positioning (direction) versus aggressive-flow clustering (magnitude) — rather than a single latent state observed twice. *[Pending: formal independence tests (distance correlation / HSIC) beyond IC-level orthogonality, on the replication month.]*

6 Spectral Localization and Efficiency Drift

Localization. Frequency-domain analysis of the level-1 imbalance process shows a red spectrum with slope -1.86 (Hurst exponent $H = 0.43$, mildly anti-persistent). The predictive content is not spread across the spectrum: essentially the *entire* directional IC is carried by the 0.005–0.1 Hz band (periods of 10–200 s). Fitting an Ornstein–Uhlenbeck process to the imbalance yields a half-life of 5–7 s, and the dominant cross-coherence between imbalance and subsequent returns sits at $\sim$68 s. A practical corollary that we verified directly: time-domain smoothing (exponentially weighted averaging) of the imbalance *reduces* IC — naive aggregation destroys exactly the band that predicts.

Drift. Between the May census window (05-19/21) and a June window (06-02/04), the 5 s IC on BTC declined from +0.45 to +0.29 — on all symbols in the same direction — while the 1 s IC was stable or improved (BTC +0.447 $\rightarrow$ +0.502). Over the same period, quoted spreads tightened from $\sim$0.1 to 0.2–0.3 bps levels shifted regime. The signal is not disappearing; its *half-life is shortening*: price discovery is accelerating as the venue matures. To our knowledge, real-time measurement of a microstructure signal's decay toward efficiency in a young venue has not been reported at this granularity. *[Pending: third time-point from the July replication month.]*

7 Robustness

Per-day rolling IC at 5 s is $+0.433 \pm 0.061$ (BTC), $+0.400 \pm 0.098$ (ETH), $+0.355 \pm 0.071$ (SOL); the bootstrap 95% confidence interval for `imbalance_qty_l1` on BTC is $[0.442, 0.465]$ (1,000 resamples). The signal is present around the clock with no dead zones (weakest 12–16 UTC, strongest 04–08 UTC) and in both volatility regimes under a median split (low-vol $+0.458/+0.435/+0.391$ vs. high-vol $+0.422/+0.394/+0.342$ across BTC/ETH/SOL). Feature rank ordering is preserved across symbols and subwindows.

We report the validation matrix honestly: BTC passes 7 of 8 cells, ETH and SOL 6 of 8. The systematic failure is temporal out-of-sample drift (IC delta -0.17 across a six-week gap, against a 0.10 tolerance) — partially attributable to the efficiency drift of Section 6, and the direct motivation for the pre-registered replication in Section 9.

8 The Execution Boundary

The census ICs are mid-price statements. The economically relevant question is what survives at the point of execution. Two boundaries are quantified.

Taker arithmetic. The expected 1–5 s move conditional on a strong signal is 0.5–2 bps; round-trip taker cost on the venue is $\sim$11 bps. Aggressive capture of the short-horizon channel is arithmetically impossible, independent of modeling choices.

Maker fills and adverse selection. The subtler boundary concerns passive execution. Table 4 conditions the 5 s IC of `imbalance_qty_l1` on fill events of a resting order at the touch (mid-cross fill model, 23 gap-audited dates, 2026-04-19 to 06-04).

Table 4: Fill-conditioning decomposition: rank IC of `imbalance_qty_l1` at 5 s under fill conditioning.

Conditioning	BTC	ETH	SOL
Unconditional	+0.453	+0.438	+0.412
Any fill event	+0.526	+0.516	+0.460
Buy fill (signal long)	+0.032	−0.061	−0.032
Sell fill (signal short)	−0.047	−0.027	−0.021

The decomposition is the point. Conditioning on the *presence* of a fill raises the IC above its unconditional level (ticks near fills carry larger imbalance). Conditioning on the *directionally consistent* fill — the one a signal-following market maker actually receives — annihilates it, on all three symbols. The mechanism: the signal says buy; the bid is placed; the fill requires the mid to fall to the bid, an adverse move; at fill time the prediction is already spent. This is adverse selection in the classical sense (Glosten and Milgrom, 1985; Kyle, 1985), made visible as a conditional-IC statement, and it locates the monetization loss at *fill selection*, not at signal decay. Fill prevalence rises from 20.8% (BTC) to 30.8% (SOL) of ticks, and profitability worsens in the same order: more fills mean more adverse selection, not more edge. The result complements Collin-Dufresne and Fos (2015): information plainly visible in the book need not be extractable through the act of trading on it.

Partial recovery by regime conditioning. Conditioning on a book-shape entropy measure recovers part of the boundary: in the low-entropy (structured-book) quintile, the imbalance IC rises from 0.45 to 0.55 (single condition) and to 0.63–0.67 in combination — a +22% lift replicated across symbols. Structured books appear to correspond to states where resting-order information is less contested. *[Pending: fill-conditional version of the entropy gate on the replication month — the decisive test of whether regime conditioning survives at execution.]*

9 Out-of-Time Replication (Pre-Registered)

At the time of writing, an independent month of data (July 2026) is in final collection. The following analyses are pre-registered and will replace this section in the next version:

1. Full census re-estimation (Tables 1, 3, 2) on the new month; the temporal-OOS validation cell re-scored.
2. Fill-conditioning decomposition (Table 4) re-estimated on the independent window.
3. Orthogonality re-tested, including formal independence measures.
4. A third time-point for the efficiency-drift trajectory.
5. Entropy-gate lift re-estimated unconditionally and fill-conditionally.

All tests run under Benjamini–Hochberg false-discovery control across the full battery, with embargoed walk-forward splits where model fitting is involved; transaction-cost parameters come from a single audited configuration source. Kill criteria are stated in advance: a claim that fails replication is removed, not re-tuned.

10 Implications and Future Work

Two-block architectures. The factorization of Section 5 suggests forecasting and trading systems in which a directional block (imbalance family) sets the sign and a volatility block (intensity family) sets the size and the veto — e.g. reservation-price skew and spread width in an Avellaneda–Stoikov market maker (Avellaneda and Stoikov, 2008; Guéant et al., 2013), with quoting disarmed when toxicity predictors fire (Cartea et al., 2018). The microprice of Stoikov (2018) is the natural implementation of the sign block at the quote level.

The queue-priority question. The fill-conditioning collapse of Section 8 is a property of *reactive* limit orders. Whether orders placed ex ante — holding queue priority, whose fills sample noise flow rather than only adverse flow (Moallemi and Yuan, 2016) — retain positive fill-time IC is, in our view, the central open question this measurement program poses. A queue-position fill simulation on recorded raw trade data is in progress.

Execution overlays. Even absent standalone monetization, a 0.45-IC seconds-horizon predictor can time the entries of slower, independently-validated strategies, converting forecast into execution-cost reduction that fill-selection effects cannot reach (the trader chooses when to cross, never resting). Quantifying the recovered basis points per trade is left to companion work.

Efficiency dynamics. The drift result invites a market-design reading: a young venue's order book becoming informationally efficient in measurable real time. Longer histories will support formal tests of convergence rates.

11 Conclusion

The perpetual-futures order book carries two large, orthogonal predictive channels: a directional channel led by book imbalance (IC up to 0.47 at seconds horizons, universal across symbols, spectrally localized at 10–200 s periods) and a volatility channel led by arrival intensity (IC up to 0.35), with no measurable cross-talk between the two families. The directional channel is bounded at execution by adverse selection — visible as a fill-conditioning decomposition that raises IC on fill presence and annihilates it on fill direction — and partially recovered by book-entropy regime conditioning. The signal's half-life is shortening as the venue matures. What the book knows, where it knows it, and where that knowledge stops being capturable can all be measured; this paper is an attempt to do so honestly.

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A Reproducibility

- **Code revision:** git SHA `[stamp at submission]` of the research platform (Rust ingestion engine + Python analysis).
- **Census window:** 2026-05-19 to 2026-05-21; 2.17 M ticks/symbol at 100 ms cadence; 21,708 subsampled IC points per symbol; horizons 1 s–15 m.
- **Fill-conditioning window:** 2026-04-19 to 2026-06-04; 23 of 31 dates pass gap audit; mid-cross fill model at the touch.
- **Drift windows:** 2026-05-19/21 vs. 2026-06-02/04.
- **Replication month:** July 2026 (in collection at draft time); pre-registered battery in Section 9.
- **Multiplicity:** Benjamini–Hochberg FDR across each reported battery; significance $p < 0.01$ for census entries.
- **Costs:** all cost statements from a single audited fee configuration (taker round-trip $\sim$11 bps at study time).

Data samples and analysis scripts are available from the author on request.